ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

SEMESTER PROJECT SPRING 2026

BACHELOR IN MATHEMATICS

Minimax Goodness-of-fit testing in High-Dimensional models

Author:
Hamza LASRI

Supervisor:
Sven WANG

Contents

Chapter 1

Introduction

In this project we analyse a model for describing the trajectory of particles of pollutants in groundwater. A successful model for describing underground flows is given by the uncertain Darcy problem. Given a domain D such that its boundary ∂D is divided in three subsets $\Gamma_{\text{in}}, \Gamma_{\text{out}}, \Gamma_N$ such that

$$\Gamma_{\text{in}} \cup \Gamma_{\text{out}} \cup \Gamma_N = \partial D, \quad \Gamma_{\text{in}} \cap \Gamma_{\text{out}} \cap \Gamma_N = \emptyset,$$

the pressure and velocity fields p and u are given by the solution of the following Partial Differential Equation (PDE)

$$\begin{cases} u = -A\nabla p, & \text{in } D, \\ \nabla \cdot u = f, & \text{in } D, \\ p = p_0, & \text{on } \Gamma_{\text{in}}, \\ p = 0, & \text{on } \Gamma_{\text{out}}, \\ \nabla p \cdot n = 0, & \text{on } \Gamma_N, \end{cases} \tag{1.1}$$

where Γ_{in}, Γ_{out} are the inlet and outlet portions of the boundary of D, and an impermeability condition is imposed on Γ_N.

Chapter 2

Minimax optimal testing by classification for Gaussian models

2.1 Description of the problem

In this section, we are studying the results of [1] which consists of finding the minimum number of observations n in order to be able to perform a test with probability error less than $\delta \in (0, \frac{1}{2})$ to the hypotheses :

$$H_0 : P_X = P_0 \text{ vs } H_1 : \mathrm{TV}(P_X, P_0) \geq \varepsilon$$

For some $\varepsilon > 0$ and a distribution $P_0 \sim \bigotimes_{j=1}^{\infty} \mathcal{N}(0, 1)$ following a Gaussian sequence model

2.2 Sobolev ellipsoid

Definition 1. *Let $s, C > 0$. We define the Sobolev ellipsoid of smoothness s and size C as the subset of real sequences :*

$$\mathcal{E}(s, C) := \{\theta \in \mathbb{R}^{\mathbb{N}} : \sum_{j=1}^{\infty} j^{2s} \theta_j^2 \leq C\}$$

For such a sequence $\theta \in \mathcal{E}(s, C)$, define the probability distribution $\mu_\theta := \bigotimes_{j=1}^{\infty} \mathcal{N}(\theta_j, 1)$, and define the set of Gaussian sequence distributions as :

$$\mathcal{P}(s, C) := \{\mu_\theta ; \theta \in \mathcal{E}(s, C)\}$$

The study of such models is motivated by the classical white noise model [2].

Chapter 3

Optimal transport based methods in Goodness-of-Fit problems

Chapter 4

Conclusion

Bibliography

[1] Patrik R. Gerber, Yanjun Han, and Yury Polyanskiy. Minimax optimal testing by classification. In Gergely Neu and Lorenzo Rosasco, editors, *Proceedings of Thirty Sixth Conference on Learning Theory*, volume 195 of *Proceedings of Machine Learning Research*, pages 5395–5432. PMLR, 12–15 Jul 2023.

[2] Y. Polyanskiy and Y. Wu. *Information Theory: From Coding to Learning*. Cambridge University Press, 2025.